MATH 152 **EXAM III** NAME Solutions

Mrs. Bonny Tighe PART ONE

11.8-10.2 TA_____ Wed. 5/1/2024

50 points

1. (4 points each) Circle the letter for the correct answer

a) a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n+1}}{36^n (2n+1)!}$

6^{2n}

$\sum_{n=0}^{\infty} (-1)^n \frac{(\pi/6)^{2n+1} 6}{(2n+1)!}$

$6 \sin \pi/6 = 6(1/2) = 3$

A) $1/2$

B) $\sqrt{3}/2$

C) 1

D) $\sin(1/2)$

E) Some other answer

b) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n+2}$

$\lim_{n\to\infty} \left| \frac{(x-1)\cancel{(x-1)^n}}{n+3} \frac{n+2}{\cancel{(x-1)^n}} \right| = |x-1| < 1$

$-1 < x-1 < 1$

$0 < x < 2$

A) $(0,2)$

B) $(0,2]$

C) $[0,2]$

D) $[0,2)$

E) Some other answer

c) The Maclaurin series representation for $f(x) = \cos(3x^2)$ is:

$\sum_{n=0}^{\infty} (-1)^n \frac{(3x^2)^{2n}}{(2n)!}$

$\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{4n}}{(2n)!}$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n 3x^{4n+2}}{2n}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{2n+1}}{(2n)!}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{4n+2}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{4n}}{(2n)!}$

E) Some other answer

d) Which of the following parabola has its vertex at (0, 0) and the directrix of $x = 2$

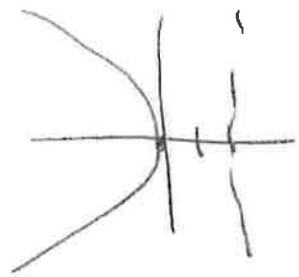
A) $x^2 = 8y$

B) $x^2 = -8y$

C) $y^2 = 8x$

D) $y^2 = -8x$

E) Some other answer



e) The second derivative of the parametric equation $x = 2 - t^2$, $y = 3 + 3t$ is:

A) $6/t^4$

B) $-3/t^4$

C) $t^3/6$

D) $6/t^3$

E) Some other answer

$dy/dx = \frac{3}{-2t} = -\frac{3}{2}t^{-1}$

$\frac{d(dy/dx)}{dx} = \frac{\frac{3}{2}t^{-2}}{-2t} = -\frac{3}{2t^2} \cdot \frac{1}{-2t}$

$+\frac{3}{+4t^3}$

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

E) Some other answer

2. (2 points each) True or False

a)If the parametric curve $x = f(t)$, $y = g(t)$, satisfies $f'(2) = 0$, then it has a horizontal tangent when $t = 2$. False

b) The Maclaurin series for $e^x = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$ False

c) The asymptotes for the hyperbola $\frac{(y-1)^2}{16} - \frac{(x+2)^2}{9} = 1$ are $y = \pm\frac{4}{3}x$ False

d) Given $f(2) = 1$, $f'(2) = 2/3$, $f''(2) = -2\cdot5/3^2$, $f'''(2) = 2\cdot5\cdot8/3^3$, etc.

then $f^{(n)}(2) = \frac{(-1)^n(2\cdot5\cdot8\cdots(3n-1)}{3^n}$ True

e) If $-2 < x < 4$ then $|x-2| < 2$ False $-4 < x-2 < 2$

3. (~~10~~ 8 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \frac{1}{\sqrt{1-x^3}} = (1+(-x^3))^{-1/2}$ $\sum_{n=0}^{\infty} \square \frac{(-x^3)^n}{n!}$

$k = -1/2$

$\square = 1 - 1/2 - 1/2(-3/2) - 1/2(-3/2)(-5/2) - 1/2(-3/2)(-5/2)(-7/2)$

$= 1 - \frac{1}{2} + \frac{1\cdot3}{2^2} - \frac{1\cdot3\cdot5}{2^3} + \frac{1\cdot3\cdot5\cdot7}{2^4}$

$n=0$ $n=1$ $n=2$ $n=3$ $n=4$

$\frac{\cancel{(-1)^n}(1\cdot3\cdot5\cdots(2n-1))}{2^n} \cancel{(-1)^n} \frac{x^{3n}}{n!}$

$1 + \sum_{n=1}^{\infty} \frac{(1\cdot3\cdot5\cdots(2n-1))}{2^n n!} x^{3n}$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(\pi/6)(x-\pi/6)^n}{n!}$$

4. (~~18~~ 13 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \cos x$ centered at $a = \pi/6$.

5 pts

$f(x) = \cos x$ $\quad f(\pi/6) = \sqrt{3}/2$

$f'(x) = -\sin x$ $\quad f'(\pi/6) = 1/2$

$f''(x) = -\cos x$ $\quad f''(\pi/6) = -\sqrt{3}/2$

$f'''(x) = \sin x$ $\quad f'''(\pi/6) = 1/2$

$$T_3(x) = \frac{\sqrt{3}}{2} - \frac{1}{2}(x-\pi/6) - \frac{\sqrt{3}}{2}\frac{(x-\pi/6)^2}{2!} + \frac{1}{2}\frac{(x-\pi/6)^3}{3!}$$

4 pts b) Use it to approximate $\cos(31\pi/180)$.

$$\cos\left(\frac{31\pi}{180}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{\pi}{180}\right) - \frac{\sqrt{3}}{2}\frac{(\pi/180)}{2} + \frac{1}{2}\frac{(\pi/180)^3}{3!}$$

$f''''(x) = \cos x$

4 pts c) How accurate is this approximation when $\pi/9 < x < 2\pi/9$? $\quad \pi/9 - \pi/6 < x - \pi/6 < \frac{2\pi}{9} - \pi/6$

If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$ $\quad M = 1$

$$\frac{2\pi}{18} - \frac{3\pi}{18} < x - \pi/6 < \frac{4\pi}{18} - \frac{3\pi}{18}$$

$$|x - \pi/6| < \pi/18$$

$$R_3(x) = \frac{1(\pi/18)^4}{4!}$$

MATH 152 **EXAM IIIA** NAME Solutions

Mrs. Bonny Tighe PART ONE

11.8-10.2 TA_____ Wed. 5/1/20224

50 points

1. (4 points each) Circle the letter for the correct answer

a) The Maclaurin series representation for $f(x) = x\arctan(x^3)$ is:

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+2}}{(2n+1)!}$

B) $\sum_{n=0}^{\infty} \frac{x^{6n+4}}{2n+1}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+4}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+4}}{2n+1}$

E) Some other answer

$x\sum_{n=0}^{\infty} \frac{(-1)^n (x^3)^{2n+1}}{2n+1}$

$\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+4}}{2n+1}$

$\frac{x^{6n+3+1}}{2n+1}$

b) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{3^{2n}(2n)!}$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

$\sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!}$

$\cos \pi/3 = 1/2$

c) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{2^n x^n}{5^{n+1}}$

A) $2/5$

B) $(-\infty, \infty)$

C) $5/2$

D) $\{5\}$

$r = \left|\frac{2}{5}x\right| < 1$

$|x| < \frac{5}{2}$

E) Some other answer

d) Which of the following parabola has its vertex at (0, 0) and the directrix of $y = 1$

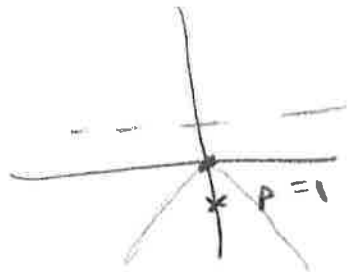
A) $x^2 = 4y$

B) $x^2 = -4y$

C) $y^2 = 4x$

D) $y^2 = -4x$

E) Some other answer



e) The second derivative of the parametric equation $x = t^2 + 1, \quad y = 1 - t^3$

A) $-3t/2$

B) $3t/2$

C) $-2/3t$

D) $2t/3$

E) Some other answer

$dy/dx = \frac{-3t^2}{2t} = -\frac{3}{2}t$

$\frac{d(dy/dx)}{dx} = \frac{-3/2}{2t} = \frac{-3}{4t}$

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

2. (2 points each) True or False

a) If the parametric curve $x = f(t)$, $y = g(t)$, satisfies $f'(1) = 0$, then it has a horizontal tangent when $t = 1$. False

b) The Maclaurin series for $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n}$ False

c) The asymptotes for the hyperbola $\frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$ are $y = \pm\frac{3}{2}x$ False

d) Given $f(2) = 1$, $f'(2) = 2/3$, $f''(2) = -2\cdot5/3^2$, $f'''(2) = 2\cdot5\cdot8/3^3$, etc.

then $f^{(n)}(2) = \frac{(-1)^n(2\cdot5\cdot8\cdots(3n-1)}{3^n}$ False $1+\frac{2}{3}$ -

e) If $-2 < x < 5$ then $|x-2| < 3$ False

$-4 < x-2 < 3$ - 4

3. (8 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \frac{1}{(1-x^3)^2}$ $(1+(-x^3))^{-2}$ $\sum_{n=0}^{\infty} \square \frac{(-x^3)^n}{n!}$

$\square = 1 + -2 - 2(-3) - 2(-3)(-4) - 2(-3)(-4)(-5)$

$\square = \underset{n=0}{1} \underset{n=1}{-2} \underset{n=2}{+3!} \underset{n=3}{-4!} \underset{n=4}{+5!}$ $(-1)^n (n+1)!$

$\sum_{n=0}^{\infty} \frac{(-1)^n (n+1)!(-1)^n x^{3n}}{n!}$

$\boxed{\sum_{n=0}^{\infty} (n+1) x^{3n}}$

$\sum^{\infty} \frac{\square(-x^3)^n}{n!}$

$(1+(-x^3))^{-2}$

$\square = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)x^3}{3!}$

4. (~~10~~ 13 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \sin x$ centered at $a = \pi/6$.

5 pts

$f(x) = \sin x \quad f(\pi/6) = 1/2 \qquad f'(x) = \cos x \quad f'(\pi/6) = \sqrt{3}/2$

$f''(x) = -\sin x \quad f''(\pi/6) = -1/2 \qquad f'''(x) = -\cos x \quad f'''(\pi/6) = -\sqrt{3}/2$

$$T_3(x) = \frac{1}{2} + \frac{\sqrt{3}}{2}(x - \pi/6) - 1/2 \frac{(x-\pi/6)^2}{2!} - \frac{\sqrt{3}}{2}\frac{(x-\pi/6)^3}{3!}$$

4 pts b) Use it to approximate $\sin(31\pi/180)$.

$$\sin(31\pi/180) = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(\frac{\pi}{180}\right) - \frac{1}{2}\frac{(\pi/180)^2}{2!} - \frac{\sqrt{3}}{2}\frac{(\pi/180)^3}{3!}$$

4 pts c) How accurate is this approximation when $\pi/9 < x < 2\pi/9$? $\frac{2\pi}{18} - \frac{3\pi}{18} < x - \pi/6 < \frac{4\pi}{18} - \frac{3\pi}{18}$

If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

$|x - \pi/6| < \pi/18$

$M = 1$

$$R_3(x) \le \frac{1\,(\pi/18)^4}{4!}$$

MATH 152 **EXAM III** NAME Solutions

Mrs. Bonny Tighe PART TWO

11.8-10.2 TA_____ Wed.5/1/2024

50 points

1. (3 points) Find the vertex, focus and directrix. Sketch the graph.

$(x-1)^2 = -(y+3)$

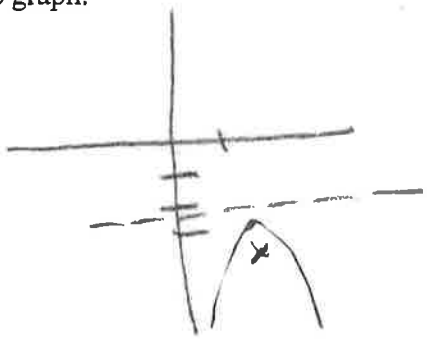
$4p = -1$

$p = -1/4$

vertex $(1,-3)$

focus $(1, -3 - 1/4)$

directrix $y = -2\frac{3}{4}$



2. (4 points) Find the center, foci, and vertices. Sketch the graph

$x^2 + 2x + 9y^2 - 36y + 28 = 0$

$x^2+2x+1 + 9(y^2-4y+4) = -28+1+36$

$(x+1)^2 + 9(y-2)^2 = 9$

$\frac{(x+1)^2}{9} + \frac{(y-2)^2}{1} = 1$

center $(-1,2)$

foci $(-1 \pm 2\sqrt{2}, 2)$

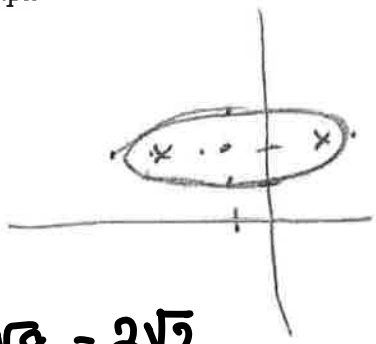
vertices $(-1 \pm 3, 2)$

$c^2 = a^2 - b^2$ $\sqrt{8} = 2\sqrt{2}$

$9 - 1$

$c^2 = a^2 - b^2$

$c = \sqrt{8} = 2\sqrt{2}$



3. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(x+2)^2 - 9(y-1)^2 = 36$

$\frac{(x+2)^2}{9} - \frac{(y-1)^2}{4} = 1$

$c^2 = a^2 + b^2$

$9 + 4 = 13$

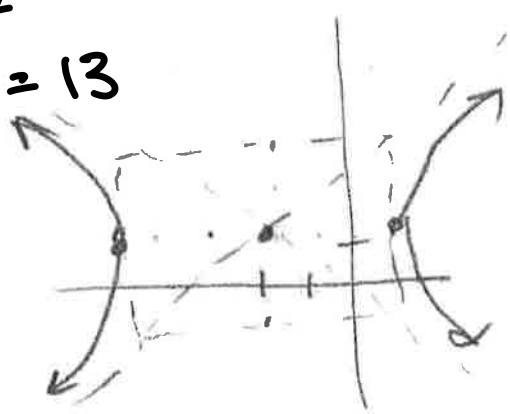
center $(-2,1)$

foci $(-2 \pm \sqrt{13}, 1)$

vertices $(-2 \pm 3, 1)$

asymptotes

$(y-1) = \pm \frac{2}{3}(x+2)$

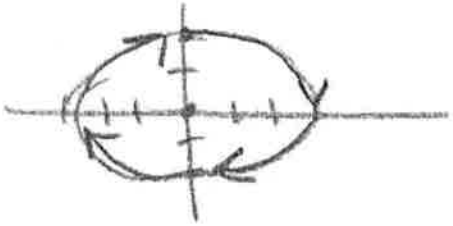


$c^2 = a^2 + b^2$

$c = \sqrt{13}$

4. (4 points) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\sin\theta, \quad y = 2\cos\theta$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$



b) (5 points) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

$$\int_0^{\pi/2} 2\cos\theta(3\cos\theta)\,d\theta = 6\int_0^{\pi/2}\cos^2\theta\,d\theta$$

$$3\int_0^{\pi/2}(1+\cos 2\theta)\,d\theta = 3\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/2}$$

$$3\left(\frac{\pi}{2} + \frac{1}{2}\sin\pi\right) - (0) = 3\pi/2$$

5. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty}\frac{(-1)^n(x-2)^n}{3^n(n+2)}$$

$$\lim_{n\to\infty}\left|\frac{(x-2)(x-2)^n}{3\cdot 3^n(n+3)}\cdot\frac{3^n(n+2)}{(x-2)^n}\right| =$$

$$\lim_{n\to\infty}\left|\frac{x-2}{3}\left(\frac{n+2}{n+3}\right)\right| = \left|\frac{x-2}{3}\right| < 1$$

$|x-2| < 3$

$-3 < x-2 < 3$

$-1 < x < 5$

Test end points

$x = -1$

$$\sum_{n=0}^{\infty}\frac{(-1)^n(-3)^n}{3^n n+2} = \sum_{n=0}^{\infty}\frac{(-1)^n(-1)^n 3^n}{3^n(n+2)} = \sum_{n=0}^{\infty}\frac{1}{n+2}$$ divergent

$$\sum_{n=0}^{\infty}\frac{(-1)^n 3^n}{3^n(n+2)} = \sum_{n=0}^{\infty}\frac{(-1)^n}{n+2}$$ Conditionally converges

$R = 3$

$I = (-1, 5]$

6. (5 points) Use series to evaluate the limit $\lim_{x\to 0}\frac{x-\sin(x)}{1-e^{x^3}}$

$$\sin x = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

$$e^{x^3} = \sum_{n=0}^{\infty}\frac{x^{3n}}{n!} = 1 + \frac{x^3}{1!} + \frac{x^6}{2!} + \frac{x^9}{3!}\cdots$$

$$\lim_{x\to 0}\frac{x-\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}\cdots\right)}{1-\left(1+\frac{x^3}{1}+\frac{x^6}{2}+\frac{x^9}{6}\cdots\right)} = \lim_{x\to 0}\frac{\frac{x^3}{6}-\frac{x^5}{5!}+\frac{x^7}{7!}}{-\frac{x^3}{1}-\frac{x^6}{2}-\frac{x^9}{6}}$$

$$\lim_{x\to 0}\left(\frac{\frac{1}{6}-\frac{x^2}{5!}+\frac{x^5}{7!}\cdots}{-1-\frac{x^3}{2}-\frac{x^6}{6}\cdots}\right) = -1/6$$

7. (10 points) Find the Taylor series for $f(x)=\frac{1}{(2+x)^2}$ centered at

a = 2

$$\sum_{n=0}^{\infty} f^{(n)}(2)\frac{(x-2)^n}{n!}$$

$f(x) = (2+x)^{-2}$ | $f(2) = \frac{1}{4^2}$

$f'(x) = -2(x+2)^{-3}$ | $f'(2) = \frac{-2!}{4^3}$

$f''(x) = 3\cdot 2(x+2)^{-4}$ | $f''(x) = \frac{3!}{4^4}$

$f'''(x) = -4!\,(x+2)^{-5}$ | $f'''(x) = -4!/4^5$

$f''''(x) = 5!\,(x+2)^{-6}$ | $f''''(x) = 5!/4^6$

$$\underset{n=0}{\frac{1}{4^2}} - \underset{n=1}{\frac{2!}{4^3}} + \underset{n=2}{\frac{3!}{4^4}} - \underset{n=3}{\frac{4!}{4^5}} + \underset{n=4}{\frac{5!}{4^6}} \qquad \frac{(-1)^n(n+1)!}{4^{n+2}} \text{ or } 2^{(2n+4)}$$

$$\sum_{n=0}^{\infty}\frac{(-1)^n(n+1)\,n!}{4^{n+2}}\frac{(x-2)^n}{n!} = \sum_{n=0}^{\infty}\frac{(-1)^n(n+1)(x-2)^n}{4^{n+2}}$$

$$\text{or } \sum_{n=0}^{\infty}\frac{(-1)^n(n+1)(x-2)^n}{2^{2n+4}}$$

8. (5 points) Evaluate the integral as a power series and approximate to T_7.

$$\int_0^{0.5} \frac{x^2}{4-x^2}\,dx = \int_0^{0.5} \frac{x^2/4}{1-\frac{x^2}{4}}\,dx = \int_0^{0.5} \sum_{n=0}^{\infty} \frac{x^2}{4}\left(\frac{x^2}{4}\right)^n dx$$

$$\int_0^{0.5} \sum_{n=0}^{\infty} \frac{x^{2n+2}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{x^{2n+3}}{(2n+3)\,4^{n+1}}\Bigg]_0^{0.5}$$

$$\frac{x^3}{3\cdot 4} + \frac{x^5}{5(4^2)} + \frac{x^7}{7(4^3)}\Bigg]_0^{0.5}$$

$$\left(\frac{(0.5)^3}{12} + \frac{(0.5)^5}{5(16)} + \frac{(0.5)^7}{7(64)}\right) + 0$$

9. (10 points) Compute the surface area obtained by rotating the given curve about the

y-axis $\quad x = e^t - t, \quad y = 4e^{t/2}, \quad 0 \le t \le 1 \qquad S = 2\pi \int_\alpha^\beta radius\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$

$dx/dt = e^t - 1 \qquad dy/dt = 4e^{t/2}(1/2)$

$2e^{t/2}$

$(dx/dt)^2 = (e^t-1)^2 \qquad (dy/dt)^2 = 4e^t$

$e^{2t} - 2e^t + 1$

one fraction

$$S = 2\pi \int_0^1 (e^t - t)\sqrt{e^{2t} - 2e^t + 1 + 4e^t}\,dt =$$

$$S = 2\pi \int_0^1 (e^t - t)\left(\sqrt{(e^t+1)^2}\right) dt = 2\pi \int_0^1 (e^t - t)(e^t + 1)\,dt$$

$$= 2\pi \int_0^1 (e^{2t} + e^t - te^t - t)\,dt = 2\pi\left[\frac{1}{2}e^{2t} + e^t - \frac{1}{2}t^2 - te^t - e^t\right]$$

$u = t \quad dv = e^t\,dt$

$du = dt \quad v = e^t$

$$2\pi\left[\frac{1}{2}e^{2t} + 2e^t - \frac{1}{2}t^2 - te^t + e^t\right]_0^1$$

$$2\pi\left[\left(\frac{1}{2}e^2 + 2e - \frac{1}{2} - e\right) - \left(\frac{1}{2} + 2\right)\right]$$

MATH 152 **EXAM IIIA** NAME Solutions

Mrs. Bonny Tighe PART TWO

11.8-10.2 TA_____ Wed.5/1/2024

50 points

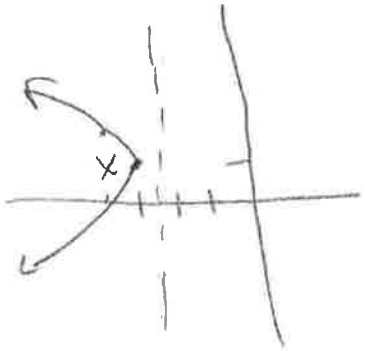
1. (3 points) Find the vertex, focus and directrix. Sketch the graph.

$(y+1)^2 = -2(x+3)$ $4p = 2$ focal width

vertex $(-3, 1)$ $p = 1/2$

focus $(-7/2, 1)$

directrix $x = -5/2$



2. (4 points) Find the center, foci, and vertices. Sketch the graph

$9x^2 - 18x + 4y^2 + 8y = 23$

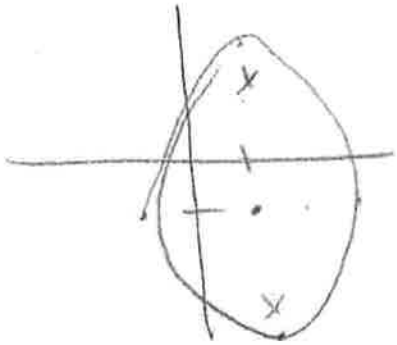
$9(x^2 - 2x + 1) + 4(y^2 + 2y + 1) = 23 + 9 + 4$

$9(x-1)^2 + 4(y+1)^2 = 36$

$\boxed{\frac{(x-1)^2}{4} + \frac{(y+1)^2}{9} = 1}$

center $(1, -1)$ vertices $(1, -1 \pm 3)$

foci $(1, -1 \pm \sqrt{5})$



3. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(x+2)^2 - 25(y-1)^2 = 100$

$\frac{(x+2)^2}{25} - \frac{(y-1)^2}{4} = 1$

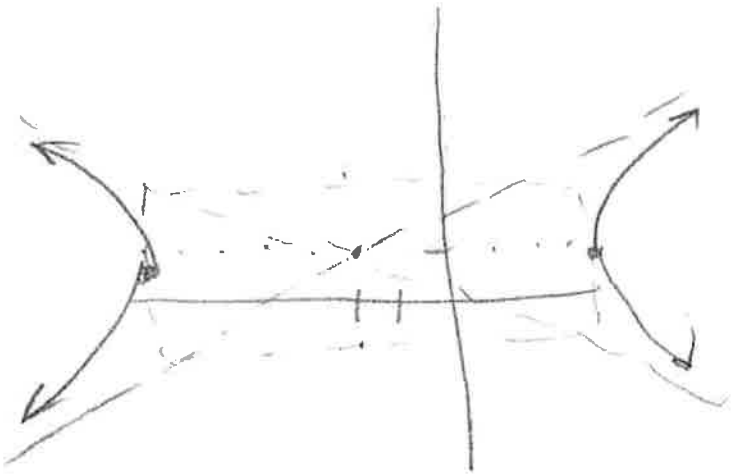
center $(-2, 1)$

foci $(-2 \pm \sqrt{29}, 1)$

vertices $(-2 \pm 5, 1)$

asymptotes

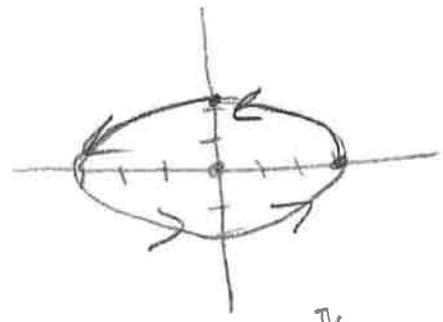
$(y-1) = \pm \frac{2}{5}(x+2)$



4. (4 points) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\cos\theta, \quad y = 2\sin\theta$

$\frac{x^2}{9} + \frac{y^2}{4} = 1$

center $(0,0)$



b) (5 points) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

$\int_0^{\pi/2} x\,dy$

$\int_0^{\pi/2} 3\cos\theta\,(2\cos\theta)\,d\theta$

$6\int \cos^2\theta\,d\theta = 3\int(1+\cos 2\theta)\,d\theta = 3\theta + \frac{3}{2}\sin 2\theta \Big|_0^{\pi/2}$

$\left(3(\pi/2) + \frac{3}{2}\sin\pi\right) - 0 = 3\pi/2$

5. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{n+2}$$

$\lim_{n\to\infty} \left| \frac{(x-2)(x-2)^n}{n+3} \cdot \frac{n+2}{(x-2)^n} \right| = \left| (x-2)\frac{n+2}{n+3} \right|$

$= |x-2| < 1 \qquad -1 < x-2 < 1 \qquad 1 < x < 3$

$R = 1$

$I = (1, 3]$

test end points

$x = 1$

$\sum_{n=0}^{\infty} \frac{(-1)^n(-1)^n}{n+2} = \frac{1}{n+2}$

diverges

$x = 3$

$\sum_{n=0}^{\infty} \frac{(-1)^n(1)^n}{n+1} = \frac{(-1)^n}{n+1}$

Cond converges

$(2x)^{2n}$

6. (5 points) Use series to evaluate the limit $\lim_{x\to 0}\dfrac{1+x-\cos(2x)}{e^x-1}$

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} x^{2n}}{(2n)!} = 1 - \frac{4x^2}{2!} + \frac{2^4 x^4}{4!} + \frac{2^6 x^6}{6!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \cdots$$

$$\lim_{x\to 0} \frac{1+x-\left[1-\frac{4x^2}{2!}+\frac{2^4x^4}{4!}+\frac{2^6x^6}{6!}\right]}{\cancel{1}+x+\frac{x^2}{2}+\frac{x^3}{6}\cdots-\cancel{1}} = \lim_{x\to 0}\left(\frac{x+\frac{4x^2}{2}-\frac{2^4x^4}{4!}-\frac{2^6x^6}{6!}}{x+\frac{x^2}{2}+\frac{x^3}{6}\cdots}\right)$$

$$\lim_{x\to 0}\left(\frac{1+\frac{4x}{2}-\frac{16x^3}{4!}-\frac{2^6x^6}{6!}}{1+\frac{x}{2}+\frac{x^2}{6}\cdots}\right) = \frac{1}{1} = 1$$

7. (10 points) Find the Taylor series for $f(x) = \ln(x-2)$ centered at a = 4

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(4)(x-4)^n}{n!}$$

$n! = n(n-1)!$

$f(x) = \ln(x-2)$ $\quad f(4) = \ln(2)$

$f'(x) = (x-2)^{-1}$ $\quad f'(2) = \frac{1}{2}$

$f''(x) = -(x-2)^{-2}$ $\quad f''(2) = -\frac{1}{2^2}$

$f'''(x) = 2(x-2)^{-3}$ $\quad f'''(2) = \frac{2}{2^3}$

$f''''(x) = -3!(x-2)^{-4}$ $\quad f''''(2) = \frac{-3!}{2^4}$

$$\underset{n=0}{\ln 2} + \underset{n=1}{\frac{1}{2}} - \underset{n=2}{\frac{1}{2^2}} + \underset{n=3}{\frac{2!}{2^3}} - \underset{n=4}{\frac{3!}{2^4}} \qquad \frac{(-1)^{n+1}(n-1)!}{2^n}$$

$$\ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(n-1)!\,(x-4)^n}{2^n\, n!} = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-4)^n}{2^n(n)}$$

8. (5 points) Evaluate the integral as a power series and approximate to T_5.

$$\int_0^{0.5} \frac{x^2}{3-x}\,dx \qquad \int_0^{0.5}\left(\frac{x^2/3}{1-x/3}\right)dx = \int_0^{0.5}\sum_{n=0}^{\infty}\frac{x^2}{3}\left(\frac{x}{3}\right)^n dx$$

$$\int_0^{0.5}\sum_{n=0}^{\infty}\frac{x^{n+2}}{3^{n+1}}\,dx = \sum_{n=0}^{\infty}\frac{x^{n+3}}{3^{n+1}(n+3)}\Bigg]_0^{0.5}$$

$$\frac{x^3}{3(3)} + \frac{x^4}{3^2(4)} + \frac{x^5}{3^3(5)}\Bigg]_0^{0.5} = \boxed{\frac{(0.5)^3}{9} + \frac{(0.5)^4}{36} + \frac{(0.5)^5}{135}}$$

9. (10 points) Compute the surface area obtained by rotating the given curve about the y-axis $x = e^t + e^{-t}$, $y = 5 - 2t$, $0 \le t \le 1$

$r = x$

$$S = 2\pi\int_\alpha^\beta radius\sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$$

$dx/dt = e^t - e^{-t}$ $\quad dy/dt = -2$

$(dx/dt)^2 = e^{2t} - 2 + e^{-2t}$ $\quad (dy/dt)^2 = 4$ "one g'knote"

$$S = 2\pi\int_0^1 (e^t + e^{-t})\sqrt{e^{2t} - 2 + e^{-2t} + 4}\,dt$$

$$2\pi\int_0^1 (e^t + e^{-t})\sqrt{(e^t + e^{-t})^2}\,dt = 2\pi\int_0^1 (e^t + e^{-t})(e^t + e^{-t})\,dt$$

$$2\pi\int_0^1 (e^{2t} + 2 + e^{-2t})\,dt = 2\pi\left[\tfrac{1}{2}e^{2t} + 2t - \tfrac{1}{2}e^{-2t}\right]_0^1$$

$$2\pi\left[\left(\tfrac{1}{2}e^2 + 2 - \tfrac{1}{2}e^{-2}\right) - \left(\tfrac{1}{2} - \tfrac{1}{2}\right)\right]$$

$$2\pi\left[\tfrac{1}{2}e^2 + 2 - \tfrac{1}{2e^2}\right]$$